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AImagery: A Multisensory Approach to Anxiety Reduction with AI, Olfactory Stimuli, and Biofeedback-Enhanced Guided Imagery

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***Abstract*—We present *AImagery*, an AI-powered immersive relaxation experience tailored to user preferences and physiological feedback. In a study with 32 participants, half of them experienced a multisensory experience with scent, biofeedback, and a personalized AI audio story based on their heart rate, self-reported mood and custom scenery. The experimental group showed a significant anxiety reduction for those with moderate to high anxiety, as opposed to the control group (non-guided meditation/baseline resting control condition). User feedback was positive and received high ratings for enjoyment, immersion and, to a lesser extend, sleepiness. Our results highlight the potential of multisensory AI-driven relaxation tools for those with elevated anxiety.**

***Index Terms*—AI, well-being, Large Language Models, multimodal, affective computing, meditation, health**

I. Introduction

Numerous studies have underscored the positive impacts of mindfulness on cognitive functions, emotional regulation, and stress reduction [1]. Mindfulness-based interventions as well as other active treatments such as relaxation and imagery have been shown to improve the clinical outcomes for patients with anxiety and depression [2]. Furthermore, the incorporation of these techniques into modern therapeutic and intervention strategies has shown promising outcomes, particularly in areas such as behavior change and cognitive restructuring. More recently, online mindfulness-based interventions have regained attention due to their scalability and accessibility, which have been associated with lower perceived stress and higher work engagement in both cross-sectional and longitudinal analyses [3]. However, despite the growing recognition and popularity of these practices for their myriad of benefits, a considerable number of individuals remain disengaged. Several reasons contribute to this, ranging from time constraints, lack of accessibility, proper guidance, to cultural and personal beliefs. In addition, the lack of personalization in such exercises may not cater to the specific needs and preferences of individuals. This problem has been pointed out by a number of research papers and reviews, which address the need for individualized or tailored interventions in various areas of health and well-being, including relaxation techniques. Stressors and coping mechanisms differ amongst individuals and across time [4]; while some may find solace in the sound of rain, others may prefer a narrative that revolves around a calm beach or a scent from their childhood memories. The 'one-size-fits-all' nature of many relaxation techniques might not appeal to everyone. Our work aims to help in recognizing the individual differences in stressors and coping mechanisms to create interactive, personalized relaxation and mindfulness experiences that do not require a conscious effort or cognitive resources to create mental scenarios or stories.

For this reason, we created *AImagery*, which explores the infusion of Large Language Models (LLMs) with traditional mind-body techniques such as biofeedback, guided imagery, and olfactory stimuli. Inspired by these techniques, we developed a multi-sensory experience uniquely crafted for each user, that provides real-time biofeedback and creates a story according to their heart rate, self-reported mood, chosen scenery, and selected odor (lavender). A randomized experiment with 32 participants was designed to compare the usability and effects on physiology and anxiety of such a multisensory system compared to a baseline resting control condition. Our findings highlight a significant drop in anxiety for those using *AImagery* with moderate anxiety scores of 43 at pre-treatment, which transitioned to non-clinical anxiety scores of 35 post-treatment (a score of 38 is often used as a cut point to detect clinically significant symptoms). This effect is noteworthy when compared with a comprehensive meta-analysis of 13 studies on mindfulness-based therapies [2], which show an average drop of 3 points as opposed to 8 points with *AImagery* (43 pre-treatment and 40 post-treatment) and approximately one point in our control condition.

Drawing from a thorough examination and rich traditions of mind-body practices, the creation of *AImagery*, complemented by a detailed user study and analytical insights, lays the groundwork for further investigation into the variables contributing to multi-sensory approaches to anxiety reduction. By bridging the gap between traditional practices and LLMs, this work aims to advance personalized, non-pharmacological interventions for anxiety reduction.

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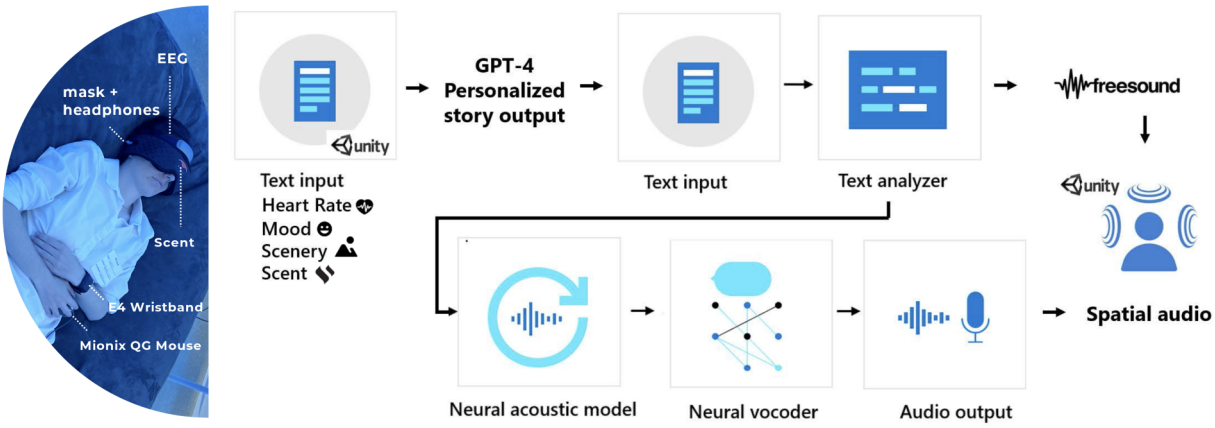

Fig. 1. *AImagery* system workflow: multi-modal user signals such as heart rate, self-reported mood, preferred scenery or memory from their past and type of scent are used as input to a large language model to create a personalized interactive experience for relaxation using AI-driven spatial audio effects as output.

II. Related Work

Previous work has highlighted the opportunities and challenges associated with mindfulness research in HCI [5], [6], as well as the integration of AI into various application domains and the potential impact on user interfaces and interactions [7]. The combination of AI-driven tools with relaxation techniques presents a novel intersection and set of challenges of two traditionally separate domains. In the subsequent sections, we discuss existing literature in both areas and how our system integrates them.

A. Mind-body techniques

Mindfulness originates from ancient Eastern traditions and has gathered especial recognition in the west due to its substantial impact on mental well-being [8]. Research has shown that regular meditation practice appears to dynamically influence brain activity measured by EEG and may enhance emotional regulation capabilities [9].

1) Guided Imagery and relaxation techniques: In a parallel vein, relaxation exercises, often incorporating Guided Imagery (GI), have been the subject of extensive investigation, consistently demonstrating their efficacy in promoting relaxation, helping with sleep, uplifting mood, and mitigating stress [2]. This type of relaxation technique has been been used for centuries and is common in complementary health approaches both in clinical care and private practices. It has been shown to reduce Post-traumatic stress disorder (PTSD) symptoms, as well as lowering stress, pain [10], depression [11], medication use [12] and many other beneficial documented results [13].

2) Effectiveness of mindfulness-based therapies: Compared to visualization or meditation, GI is often directed, either by an instructor, video, audio recording, written script or using one's imagination. Individuals internally visualize peaceful scenes, memories, or places, engaging all senses to create a multisensory experience that promotes relaxation and calmness. This immersive process can mentally transport someone to a serene space, providing comfort and tranquility. Supporting this, extensive research, including meta-analyses on mindfulness-based therapies, have highlighted the effectiveness of practices like GI in managing anxiety [2]. In particular, the results of 13 studies revealed that individuals with non-clinical anxiety demonstrated significant reductions in the State-Trait Anxiety Inventory (STAI) scores from pre-treatment mean (35.91) to post-treatment (31.25) and follow-up (29.35). STAI scores have been previously defined as "no or low anxiety" (<37), "moderate anxiety" (38-44) and "high anxiety" (>45). A cutoff score between 38 [14] and 40 [15] is most commonly used to define clinically significant symptoms, which is considered when a patient no longer meets the diagnostic criteria for the disorder. Furthermore, those with moderate clinical anxiety at pre-treatment (42.94) transitioned to non-clinical levels post-treatment (39.73) and maintained mild levels at follow-up (40.33). Even individuals with high clinical anxiety at the onset (52.87) experienced reductions to moderate levels post-intervention (47.20) and during follow-up (46.54). This substantial decrease not only underscores the therapeutic potential of interventions like GI but also paves the way for enhancing these techniques with additional modalities.

3) Incorporation of additional sensory modalities: Incorporating 'Healing Touch' into GI sessions has been shown to further alleviate PTSD symptoms, surpassing the efficacy of standard treatments [16]. Similarly, the integration of GI with music, has demonstrated promising benefits in improving psychological health for clinical populations, including those who have survived trauma [17]. Although there is a lack of research incorporating olfactory feedback with GI, the historical use of aromatic scents and incense in meditative practice underscores their calming properties. Scent has been shown to aid in anxiety reduction [18], [19], improve mood [20], [21], sleep [22] and overall well-being [23].

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4) Biofeedback and combined interventions: The incorporation of additional mind-body techniques like biofeedback further underscores the multifaceted approach to anxiety reduction, as evidenced by its effectiveness in enhancing patient recovery post-surgery [24]. More specifically, researchers found that GI reduced anxiety and perceived pain, while biofeedback showed a steady decline in stress and medication use. Beyond these, the combined intervention of GI and biofeedback yielded the most significant outcomes.

In summary, the integration of additional sensory modalities and mind-body techniques into guided imagery presents a significant opportunity for enhancing its therapeutic efficacy. However, while these techniques have proven effective, they typically adopt a standardized approach that may not accommodate the distinct preferences and requirements of each person, potentially impeding the full realization of their benefits [4]. Consequently, our objective in this work was to design a deeply personalized and holistic experience that not only builds upon the foundational strengths of traditional mind-body practices but also utilizes the innovative capabilities of the latest large language models to create an interactive experience tailored to the individual preferences and emotional state.

B. HCI-AI approaches to anxiety reduction

Historically, the application of AI in mental health has centered on identification and diagnosis of disorders, complemented by the development of therapeutic chatbots and applications. The intersection of AI and therapy is not a recent phenomenon, with the inception of the first chatbot, ELIZA, tracing back to the 1960s.

1) Machine Learning in mental health: In contemporary research, a substantial body of literature encompassing hundreds of studies examines the role of machine learning (ML) in mental health contexts [25]. Concurrently, the emergence of dozens of ChatGPT therapist bots and AI apps to relieve mental health problems are growing (e.g, ThoughtCoach, SommerAI, Mindsum Bot, JungGPT, Lotus). GPT has already shown promise in various health domains [26], spanning diagnosis, therapeutic interventions, research, and clinical operations.

2) Immersive and interactive systems: It's clear that ample opportunities exist to expand ML applications in various facets, especially with the integration of interactive and immersive systems. In this paper, we explore the use of GPT with a variety of inputs such as individual preferences, emotional state and physiology, to craft a more immersive and effective relaxation session. By engaging multiple senses at once, multisensory experiences have been shown to be more immersive and effective than unisensory experiences [27]. By integrating sound, scent, and narratives, multisensory relaxation techniques can provide a more holistic and effective approach for anxiety reduction.

3) 3D Audio and Visual Immersion: The use of 3D audio or soundscapes [28], [29], can further enhance the sense of immersion and reflection, as found in studies exploring the autonomous sensory meridian response (ASMR) [30]. Similarly, the use immersive visuals, such as in Virtual Reality (VR) or Mixed Reality (MR) with biofeedback, enhances immersion, boosts mindfulness and relaxation [31]–[34].

4) AI and Biofeedback Mechanisms: In addition to these, the combination of AI with biofeedback mechanisms [35], [36] offers a tailored experience that can adapt to the user's current state, enhancing the relaxation efficacy.

In summary, with the increasing digitization of relaxation techniques, novel technologies and multiple sensory inputs/outputs [37], [38] that can be used during day and night [39], there is a growing number of multisensory interventions and systems that could be further studied. These systems must strike a balance that we aim to address in this paper: they should be user-friendly, engaging, and effective, while also addressing ethical dilemmas and cultural perspectives related to machines understanding implicit signals and emotions [40].

III. System Description

AImagery consists of (i) AI guided imagery, (ii) auditory and olfactory output, and (iii) real-time biofeedback.

A. AI-Assisted Guided Imagery

We developed an open-sourced [1] Unity3D application with attached scripts written in C# that consists of three main components: 1) Story, 2) Sound, 3) Audio 3D FX. First, the Story component, combines the user-provided feelings, desired scenery, heart rate (HR), and some additional information into a prompt (see Appendix for the prompt). This information is then processed by GPT-4, which generates an output text with the personalized story along with three sound tags associated with the most related background sound effects to the generated story. Next, the Sound component uses Freesound's public API to search for sounds with the tags generated. If no sounds exist with all of the tags, the last tag is discarded and the search is repeated for the remaining tags. This process continues until a match is returned, and that sound is played to the user in the background in a continuous loop. Lastly, the Audio 3D FX component uses text-to-speech to generate a narrative voice for a 10 minute story. We use the Jane Neural voice from Azure Cognitive Services–a neutral female voice with United States accent– (see more details about the custom Speech Synthesis Markup Language (SSML) that we use in the Appendix). Following pilot sessions and user input, we adjusted the narrative delivery by slowing the prosody rate and adopting a whispering style. We introduced strategic pauses for emphasis and incorporated 3D audio to enrich the auditory experience. Additionally, we refined the storytelling prompts to exclude references to bees, addressing user preferences and avoiding unintended associations with the calming scent of lavender and the buzzing of bees. Each story is uniquely generated for each user. See the Additional Materials to listen and read the stories generated for each participant. A subset of the stories can be found in the Appendix and Additional Materials, with an example of what a non-personalized GPT prompt would give as output.

[1] https://github.com/microsoft/AImagery

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B. Real-time Biofeedback

We collect real-time HR from the Mionix Naos QG and process this information in Unity3D. Our C# code is based on the WebSockets Mionix BETA API. Once HR information is obtained, we transform its frequency to audio stimuli by playing the sound of a heart beat concordant with the users' real-time HR.

C. Auditory and Olfactory Stimuli

Auditory stimuli from the Unity app is streamed via Bluetooth to a pair of headphones that are integrated in a sleeping mask. Odor is released through the mask via an add-on round pad made of felt that contains lavender essential oil. The AI narrates a personalized story based on input information that expresses a user's state at the beginning of the experience (including their HR and self-reported mood), preferred scenery, and lavender scent. Three streams of 3D audio are dynamically generated for each user: 1) AI voice, 2) HR biofeedback and 3) scenery sounds. The 3D audio is generated in Unity using a full-sphere surround sound to create an immersive auditory experience and is streamed separately to the left and right ear. This technique positions the whispering voice as if it's emanating from behind the listener, gently oscillating between ears. This approach was inspired by other audio experiences that trigger the autonomous sensory meridian response (ASMR), enhancing the sense of presence and relaxation. Environmental sounds are located at a further distance from the subject, while the heartbeat is audibly centered at chest level, creating a more intimate and personal soundscape.

IV. Methods

A. Participants

Thirty-two individuals (18 female, 14 male) with an age range between 18–54 years participated in this study and were recruited through posters and e-mails. There were no significant differences in terms of age and gender between the two study conditions (see Appendix Figure 4). Participants were considered eligible if they were 18 years old or older and did not have odor allergies. All subjects signed a consent form before participating in the study and were compensated with a $25 voucher at the end of the experiment. The study was reviewed and approved by the Institutional Review Board.

B. Experimental design and procedure

A randomized between-subjects experiment was designed to compare the effects on physiology and anxiety of *AImagery* compared to a baseline resting control condition. The study was conducted in a quiet room, and all environmental conditions were kept as similar as possible across both the experimental and control groups. This included controlling for noise levels, lighting, room temperature, furniture and session duration. We provided standardized instructions to both groups to minimize variability in participants' experiences. All subjects used the same bean bag and were asked to lay down and close their eyes during the duration of the baseline measurements (5 minutes), followed by 10 minutes of quiet rest or active intervention.

On the day of the study and after obtaining consent for participation and confirming eligibility, the experimenter proceeded to set up the physiological sensors and test their connectivity and signal quality. Participants filled out a series of questionnaires related to their demographics, anxiety, mood and sleep, and were then guided by the experimenter to lay down on a bean bag. Once the subject was laying down, the experimenter ensured that the sensor data was streaming and had a good signal quality. The subjects were then instructed to remain quiet, still and keep their eyes open while blinking normally during 5 minutes of baseline measurements. After 5 minutes and while the participant was laying down, the experimenter read a script. If they were in the control condition, they were told to continue to lay down and to close their eyes for 10 more minutes. They were not given any specific instructions related to relaxation or imagery, nor they were exposed to any olfactory stimuli, soundscapes, or biofeedback mechanisms. In the experimental condition, participants were asked to close their eyes while an AI voice would guide them through a relaxing scenery of their choice with sounds, scent and the sound of their heart beating (see Appendix for detailed script). After the debrief, the experimenter helped the participant to put on the sleeping mask, input the users' preferences to the system, did a sound test and proceeded with the *AImagery* experience for 10 minutes. To give the participant some privacy, the experimenter left the room in both conditions. After the intervention, participants filled out the final round of questionnaires related to their experience and their perception of using AI systems to relax.

C. Data Collection

Data from 31 participants were included in the analysis (P12 dropped out during the experiment–more details are described in the Findings section). The user study primarily focused on collecting three types of data: 1) **Usability** and perception of the system, **Subjective** anxiety, and 3) **Physiological** changes with and without *AImagery*.

*1) **Usability**:* All participants filled out a series of questions related to their overall user experience and AI preferences in the context of guided imagery. Those in the experimental condition filled out some additional questionnaires related to the system, such as preferences in regards to the AI voice, HR, scent, sounds, and the perception that they had about the abilities of the AI to perceive and adjust to their emotions.

*2) **Subjective** Data:* We collected demographics and subjective self-reports of variables that might have impacted the user experience through a series of questionnaires in Qualtrics. These included questions related to their anxiety level, their likelihood to doze off in everyday life situations as well as the quality of their sleep the night prior to the study. We also collected the mental imagery capabilities [41] of participants through the Plymouth Sensory Imagery Questionnaire (Psi-Q) [42]. To assess their anxiety pre and post intervention, we used the Form Y-1 of the State Trait

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Anxiety Inventory (STAI) [43], a validated questionnaire with 20 items to report how participants were feeling at that specific moment in time. For sleep, we used the five-item Richards-Campbell Sleep Questionnaire (RCSQ) [44] and the Epworth Sleepiness Scale (ESS) [45].

3) ***Physiological*** *Data:* A wearable Muse S (Gen 2) EEG and Empatica 4 wristband were used to passively capture the physiological signals of the user throughout the duration of the experiment. Blood Volume Pulse (BVP) was collected by the E4 wristband. Brain activity data was collected from four active EEG electrodes–TP9, AF7, AF8, and TP10 (reference electrode at Fpz). Baseline HR and resting-state EEG activity was collected for 5 minutes prior to the intervention to provide a measure of brain activity when the individual was awake and not performing any assigned cognitive task. HR sampling frequency was 1Hz, except for P11 due to an issue with the sensor. We also excluded the EEG data from two participants (one in each condition) due to a failure with the sensor.

D. Data analysis

All analysis were computed using custom Python scripts. Summary statistics such as mean and standard deviation were used to summarize quantitative data from subjective questionnaires related to usability and user experience. For anxiety scores and physiological changes we conducted both parametric and nonparametric tests, depending on the nature of the data and assumptions they met. Before proceeding with parametric tests, we checked for normality using Shapiro-Wilk tests in all the analysis described below:

1) ***Subjective*** *Anxiety:* To determine if there was a significant difference between pre and post anxiety scores we performed paired t-tests for the experimental and control conditions separately. We followed the guidelines for the STAI scoring [43]: scale from 1 to 4, with 1 being "Not at all", 2 "Somewhat", 3 "Moderately So" and 4 "Very much so". Items 1, 2, 5, 8, 10, 11, 15, 16, 19, and 20 are inverted/reversed coded (positive emotion), meaning their scoring is the reverse of the 1-4 scale. We conducted two types of analysis including: 1) all participants and 2) only those who had a moderate to high anxiety score (clinically-significant symptoms).

2) ***Physiological*** *Changes:* We first applied a non-causal variable threshold (NCVT) filter to heart rate data. After extracting baseline and intervention intervals, we calculated the average for each. An outlier was statistically identified using the interquartile range (IQR) method and removed from the analysis. Paired tests (either t-test or Wilcoxon for not-normally distributed data) were conducted for within-group comparisons. We used a two-way ANOVA to assess the effects of group (AI vs Control) and time (Baseline vs Intervention) on HR. Next, we used separate paired t-tests on the AI and Control groups to compare HR during the baseline and experience intervals as well as between conditions, followed up with post-hoc independent t-test comparing the change (or delta) in HR from baseline to experience between the AI and Control groups.

We processed and analyzed the Inter-Beat Interval (IBI) data, specifically looking at the Root Mean Square of Successive Differences (RMSSD) metric, which is an indicator of Heart Rate Variability (HRV). After computing the RMSSD for baseline and during the experience samples. Paired tests (either Wilcoxon for AI (baseline) or paired t-test for the other conditions) were conducted for within-group comparisons. Finally, a post-hoc Mann-Whitney U Test was used for between-group comparisons based on the difference between intervention and baseline RMSSD values.

Finally, for brain activity, we first filtered data points to exclude eye blinks, jaw clenches, and poor electrode signal quality recorded by the EEG. In addition, we downsampled the data to approximately 128 Hz and applied a band-stop filter between 48-52 Hz to remove any electrical noise originating from AC power sources. The mean and standard deviation of the alpha and theta bands' power were computed for each electrode site. Through this preprocessing approach, we aimed to optimize the EEG data, allowing for a clearer discernment of the individual's state of alertness or relaxation.

V. Findings and Discussion

A. User Experience

Participants who were in the experimental condition reported high levels of enjoyment, relaxation, immersion, feeling refreshed and positive. This is reflected in the mean values which are equal or above 4 on a 5-point Likert scale as shown in Figure 2 and quotes like *"I usually hate meditating and do not practice it, but this felt more accessible than being told to meditate"*,*"Loved the experience overall!"* or *"It was very relaxing"*. For most variables, participants tended to agree or strongly agree, as indicated by the high percentage of responses in these categories. This suggests that the experience was generally well received. While still positive, participants' value for feeling sleepier, rested, and comfortable was lower compared to other variables, with values between 3 and 4.

1) Dropout and negative reactions: Out of the 32, there was only one participant (P12) who dropped out of the study within 2 minutes of the experience. She immediately removed the mask and described the voice as *"super creepy,"* particularly the whispering was *"too much"* and she felt very uncomfortable with the 360/immersive sound effect: *"It felt like a robot wanted to control my experience."* She also reported that *"even if it was a person whispering it would be too creepy."*.

2) Positive feedback and relaxation: Aside of P12, more than 90% of participants in the experimental condition agreed/strongly agreed about feeling more relaxed after the experience, which coincides with the HRV and STAI findings (described in the following sections). Most participants perceived the system as being designed or developed to exhibit compassionate and empathic behaviors (as shown in Figure 2). Several participants highlighted the ease of use of *AImagery* compared to traditional meditation practices. Even those who generally avoided meditation expressed a newfound affinity for this approach, emphasizing its simplicity and user-friendliness.

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3) Variety of experiences and preferences: The study revealed a wide variety of experiences and preferences among participants engaging with *AImagery*. While some aspects, like immersive environments and personalization, received unanimous praise, others, such as audio ambiguity reflected a broader spectrum of reactions. These findings underscore the need for continued refinement and customization of AI-guided relaxation experiences to cater individual preferences and sensitivities. Further research can delve deeper into the nuanced interplay between personalization, sensory stimuli, and relaxation responses, providing valuable insights into the future of AI-driven well-being interventions. In the following subsections we dwell deeper into each one of the variables.

4) Olfactory, auditory stimuli, and biofeedback: Participants valued the immersive audio qualities of the session. The integration of heart rate monitoring and spatial sound effects was very appealing to the majority of users. E.g, P3 noted that *"I loved the heart rate feature, and the background sounds (for instance, the sea sound). Also, the fact that the voice goes form one ear to the other. I felt that was very pleasant and immersive"*. While some users described the sounds as relaxing and comparable to their podcast-based relaxation routines, P6 and P11 found it disconcerting. The AI's interpretation of their selected sceneries led to diverse auditory experiences: P6's choice of a village in the south of France prompted the AI to incorporate church bell sounds, whereas P11's selection of a living room scene resulted in the playback of a ticking clock. These soundscapes, while intended to be relaxing, were perceived differently by the individuals.

Similarly, the AI voice also caused mixed feelings. Seven participants described the voice as "creepy" and suggested a more realistic or natural voice (P8), the ability to choose male or female voice, accent, etc. (P9). Others suggested changing the voice style, e.g, P11 pointed that the whisper had an *"eerie feeling to it"*, similarly P9 noted: *"The voice was uncomfortable and sent a shiver down my spine (...) but about halfway through I made an active choice to relax and that helped me fully immerse in the experience"*.

While the voice had conflicted views, the scent of lavender, infused in the AI's story, received positive feedback from all participants. For example, P10 shared: *"The lavender smell was extremely relaxing and the entire set up was ideal for relaxing"*. Similarly, P6 noted that *"The scent was nice, I felt like it [was] released from the mask whenever the AI mentioned 'smelling lavender'"*.

5) Personalization through AI: The AI's ability to create a unique experience based on users' preferred scenery and biometric data left a positive impression. Participants found the AI's interpretation of what would calm them intriguing and unique. Some believed that the incorporation of feedback from vital signs allowed the AI to adjust its tone, sounds, and guidance dynamically accordingly with their shifting emotional states: E.g, P4 noted that *"I think that the feedback from my vitals allowed the AI to self-correct on its tone, sounds, and guided settings as my feelings shifted from tense unfamiliarity of the AI's setting to distracted calm based on the AI feedback"*. P18 choose a scenery from their past and expressed that *"the AI did integrate some of my current emotions into the experience which I thought was very interesting (...) one of my current emotions was excitement for the future and desire to connect with loved ones (...) the experience talked about my family (...) talking about memories of the past and hopes for the future"*.

Despite having access solely to the user's initial self-reported mood and heart rate, the AI crafted narratives that seemed to sync with the user's emotional state dynamically. For example, "I feel my heart rate slowly dropping, now at a pace of 83 beats per minute, as the stress from my medical school applications starts to fade away".

B. Effects on Anxiety

The assessment of anxiety levels, as measured and scored following the State-Trait Anxiety Inventory guidelines is illustrated in Figure 3. The plot displays individual changes in STAI scores from pre to post intervention for all participants, including those with baseline STAI scores equal to or greater than 38 (considered moderate to high anxiety). Upon examination of the data presented in Figure 3, several key findings emerge:

1) Significant Reduction in Anxiety for AImagery Group: Among participants with baseline STAI scores equal to or greater than 38, the application of *AImagery* resulted in a statistically significant reduction in anxiety levels ($p = 0.0324$). This reduction is particularly notable as it pertains to individuals with moderate to high baseline anxiety. Our cohort also included one participant that had a STAI score of 55 (what is considered a high clinical anxiety at the onset), which decreased to levels below 40, a preliminary but significant drop compared to previous work.

2) Control Group Anxiety Reduction: The control group also experienced a significant reduction in anxiety levels ($p = 0.0425$). However, this reduction was not statistically significant for those with moderate to high anxiety levels ($p = 0.7462$).

These findings collectively suggest that AI-guided imagery interventions augmented with scent and biofeedback hold promise in effectively reducing anxiety levels, especially in individuals with elevated baseline anxiety. The control group's reduction in anxiety, while observed across the average of all participants, it does not hold among those with initially high anxiety levels. These results underscore the importance of personalizing experiences based on the user as well as the potential to further study multisensory relaxation techniques guided by AI as a potential valuable tool for anxiety management and stress reduction (as reported by a standarized psychological inventory used for diagnosis in clinical and other medical settings).

C. Effects on Physiology

1) Heart Rate: The analysis of heart rate responses to the intervention is presented in Table III, providing a comprehensive view of the statistical outcomes. The two-way ANOVA results demonstrated that there was a significant effect of time on

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heart rate, indicating that heart rate underwent changes from baseline to the intervention phase across both the control and experimental groups. However, it's important to note that no significant interaction between group and time was observed, suggesting that the nature and magnitude of these changes were not significantly different between the two groups.

A post-hoc independent t-test revealed that both the control and experimental groups exhibited a statistically significant decrease in heart rate from their respective baseline measurements to the intervention measurements. The significance of this reduction was reflected in the p-values ($p < 0.05$) for both groups. It is important to acknowledge that this decrease was observed consistently in both groups, suggesting that *AImagery* and the resting control group were comparably effective in inducing a reduction in heart rate.

2) Heart Rate Variability: A Mann-Whitney U Test for between-group comparisons based on the difference between baseline and intervention RMSSD values found that a statistically significant difference emerged between the AI and control groups (p = 0.0261, U = 58), as depicted in Figure 3. Paired tests for within-group comparisons revealed that neither the AI group nor the control group exhibited a significant difference between their pre (baseline) and post-experience (during intervention) HRV values (Wilcoxon Signed-Rank Test for AI: p = 0.2412, W = 33), T-Test for control: p = 0.1021, T = 1.7415). This indicates that HRV did not significantly change across time within either group.

In summary, when analyzing the HRV changes within each group separately, neither the AI group nor the control group exhibited a significant alteration from before to after the experience. However, when comparing the HRV changes between the two groups, a statistically significant difference emerged, suggesting that the AI group and the control group responded differently to the experience in terms of their HRV changes. The two groups displayed contrasting responses to the experience concerning HRV. The AI group appeared to benefit, as evidenced by an increase in HRV (from $\mu = 70$, std=37 to μ = 74, std = 24), whereas the control group did not exhibit this benefit and instead showed a reduction in HRV (from M = 71, std = 36 to M = 60, std = 24).

These results offer valuable insights into the functioning of the autonomic nervous system (ANS), particularly regarding the balance between the sympathetic (SNS) and parasympathetic (PNS) branches, and how systems like *AImagery* may influence this balance. HRV represents the variation in intervals between heartbeats, known as R-R intervals or beat-to-beat variability and serves as an indicator of the autonomic nervous system's (ANS) function, mirroring the interplay between the parasympathetic nervous system (PNS)—associated with relaxation—and the sympathetic nervous system (SNS)—linked to stress responses. Typically, stress is correlated with lower HRV due to increased SNS activity, whereas recovery is marked by higher HRV and PNS dominance [46]. In our research, the reduced HRV in the control condition might indicate lingering stress or reduced adaptability. In contrast, during the *AImagery* experience, the parasympathetic system, which promotes rest and digestion, appeared to dominate. This dominance was reflected in the reduced HR (indicating a relaxed state) and increased HRV (indicating heightened adaptability and resilience during relaxation).

These findings shed light on the potential therapeutic benefits of multisensory AI-guided relaxation techniques in modulating the autonomic nervous system and promoting a more relaxed physiological state.

3) Brain Activity: The results of the analysis and rationale for brain activity (specifically focusing on alpha and theta wave frequencies, with higher alpha and theta power suggestive of increased relaxation and sleepiness, respectively) is presented in the Appendix and Table IV. A within-group comparisons revealed a non-significant difference in alpha activity for both the experimental (AI) group ($p = 0.273$) and the control group ($p = 0.613$). However, significant differences emerged in theta activity for both groups, with $p < 0.001$ for the experimental group and $p = 0.004$ for the control group. Between-group comparisons during the intervention phase did not reveal significant differences in alpha ($p = 0.440$) or theta ($p = 0.293$). These findings suggest that while there were observable differences in theta activity between the baseline and intervention phases within groups, the AI intervention did not result in significant differences in brain wave activity compared to the control group.

VI. Conclusion

We described the development and evaluation of *AImagery*, an AI-driven relaxation system that uses mind-body techniques and GPT to create personalized relaxation experiences based on individual user preferences, HR and mood. The study revealed that *AImagery* significantly reduced anxiety, particularly among participants who initially reported moderate to high levels. It also improved HRV, indicating a deeper relaxation state. Despite the generally positive feedback, there are some limitation to this work, including a small sample size (which can significantly impact the experimental power of the study (N = 32), the study's laboratory environment, and single-session design, which call for further research to assess the variables influencing its efficacy. Even though previous work has combined guided imagery with music and biofeedback, the effect of odor on guided imagery alone remains unknown. Similarly, the use of AI combined with guided imagery and other mind-body techniques remains under-explored. To conclude, *AImagery* exemplifies the potential of integrating AI with mind-body practices, offering significant therapeutic benefits for those with heightened anxiety levels.

Ethical Impact Statement

The deployment of AI-driven systems like *AImagery* brings to the fore significant ethical considerations that must be carefully navigated. While the system has shown considerable promise in aiding relaxation and reducing anxiety, the immersive features have also elicited concerns among a subset of users, particularly regarding the AI-generated voice, highlighting the necessity for a nuanced approach to

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personalizing the voice and sounds generated by the AI. This underscores the broader ethical imperative that systems like *AImagery* must be designed and implemented with a foundational commitment to human values, ethics, and the end-user experience, always ensuring that the content generated is safe.

In addressing potential risks, our approach prioritized the promotion of responsible communication and the practice of informed consent. This aligns with the recommendations of prior research in the field [47]–[50]. By adhering to these established guidelines, we aim to mitigate ethical concerns and enhance the integrity of human-AI interactions.

Lastly, as we continue conducting longitudinal studies in more naturalistic setting, such as integrating AI into healthcare, it is crucial to maintain a balance between user engagement and ethical responsibility. Systems like *AImagery* must not only be user-friendly and effective but also designed with a conscientious regard for the ethical dilemmas and cultural nuances associated with machines that interpret how the human is feeling, physiology, preferences, and overall personal health information. In doing so, we can ensure that AI serves as a tool for enhancing human well-being, grounded in the principles of transparency, accountability, and fairness.

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APPENDIX

All interventions and measurements were conducted in a controlled room at the *Anonymous Center*, during July of 2023. The odor pad was replaced after each subject, and the mask, bean bag, and physiological sensors were sanitized after each use. The room was ventilated and the air was filtered.

A. EEG data

In order to objectively measure the users' relaxation levels, we focused on neural oscillations in the frequency range rhythm of 8-13 Hz (alpha) and 4-7 Hz (theta) bands, which have been historically linked with relaxation and drowsiness, respectively. Alpha waves are commonly encountered in individuals who are awake but in a tranquil and restful cognitive state, particularly when their eyes are closed or during specific meditative practices [51]. On the other hand, theta waves can be observed during meditation, daydreaming, the state of flow, or phases of drowsiness and hypnagogia [52]. Further research and analysis is recommended to explore the potential impacts of multisensory relaxation systems assisted by AI on brain wave patterns. For instance, by segmenting the EEG into non-overlapping or overlapping windows (e.g., window size = 1s, overlap = 50%), we could further investigate the temporal evolution of the alpha and theta activity. This might be particularly useful if the individual's state of relaxation or sleepiness varied during the recording. By not segmenting, we risk treating the EEG data as stationary (i.e., assuming its statistical properties do not change over time). The decision to not segment in the initial analysis was driven by our interest on the general trend and clinical benefits rather than moment-to-moment fluctuations, but future analyses might benefit from such segmentation.

STAI ANOVA Results

Source	$\sum$ of Squares	df	F	Significance
Group	0.672	1	0.008	$p = 0.928$
Time	418.081	1	5.059	$p = 0.028$*
Interaction	34.453	1	0.417	$p = 0.521$
Residual	4793.133	58		

TABLE I
TWO-WAY ANOVA WITH FACTORS GROUP AND TIME FOR STAI.

HRV ANOVA Results

Source	$\sum$ of Squares	df	F	Significance
Group	556.488	1	0.579	$p = 0.449$
Time	238.502	1	0.248	$p = 0.620$
Interaction	933.595	1	0.972	$p = 0.328$
Residual	53743.227	56		

TABLE II
HEART RATE VARIABILITY (HRV) RESULTS OF THE TWO-WAY ANOVA WITH FACTORS GROUP AND TIME.

B. Sample size and experimental power

The reduction in sample size due to the removal of outliers or filtering for moderate to high participants can significantly impact the experimental power of the study. With smaller sample sizes, the variability in the RMSSD (Root Mean Square of Successive Differences) measurements increases, which can obscure true differences between the Experimental and Control groups or between Baseline and Intervention periods. This reduction in power makes it more challenging to detect statistically significant effects, as evidenced by the mixed normality results and non-significant ANOVA findings. Given the normality results, where the baseline RMSSD for the Experimental group does not follow a normal distribution, while other conditions do, a traditional parametric ANOVA might not be appropriate due to the violation of normality assumptions. Thus, Fig.3 uses Mann-Whitney U Test and Wilcoxon Signed-Rank Test instead.

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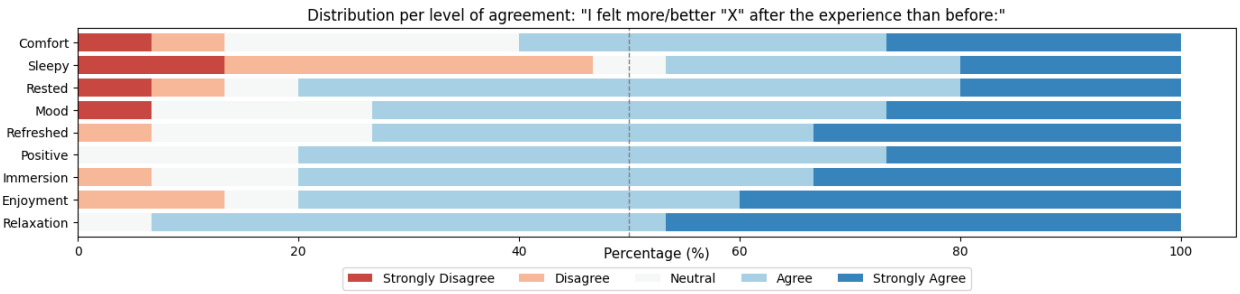

Fig. 2. The mean rating for enjoying the experience was μ=4.07, std=1.03, M=4, IQR=1 (were 5 is strongly agree). They also felt immersed (μ=4.07, std=0.88, M=4, IQR=1) and rated feeling comfortable using the AI as a guided imagery tool (μ=3.67, std=1.18, M=4, IQR=1.5). Participants felt more positive (μ=4.07, std=0.7, M=4, IQR=0.5), rested ($\bar{X}$=3.8, std=1.08, M=4, IQR=0) and were in a better mood (μ=3.87, std=1.06, M=4, IQR=1) and felt more refreshed (μ=4, std=0.93, M=4, IQR=1.5) than before the experience. On average, participants remained neutral about feeling more sleepy (μ=3.07, std=1.44, M=3, IQR=2).

HR ANOVA Results

Source	$\sum$ of Squares	df	F	Significance
Group	141.94	1	2.01	p = 0.161
Time	497.95	1	7.07	p = 0.0102*
Interaction	1.69	1	0.02	p = 0.87
Residual	3945.42	56		

Post-hoc Tests

	Control Group				Experimental Group			
Metric	**Baseline**	**Intervention**	**t-stat.**	**Sign.**	**Baseline**	**Intervention**	**t-stat.**	**Sign.**
Mean	72	67	4.586	p = 0.0003*	76	70	2.563	p = 0.0236*
Std. Deviation	8.39	6.82			7.49	8.27		

TABLE III

HEART RATE ANALYSIS: RESULTS OF THE TWO-WAY ANOVA WITH FACTORS GROUP AND TIME AND FOLLOW-UP T-TESTS.

EEG ANOVA Results

Source	$\sum$ of Squares	df	F	Significance
Alpha				
Group	0.031	1	0.894	p = 0.349
Time	0.009	1	0.268	p = 0.606
Interaction	0.002	1	0.077	p = 0.782
Residual	1.930	54		
Theta				
Group	0.018	1	0.709	p = 0.403
Time	0.174	1	6.769	p = 0.011*
Interaction	0.017	1	0.666	p = 0.417
Residual	1.388	54		

	Control Group				Experimental Group			
Metric	**Baseline**	**Intervention**	**t-stat.**	**Sign.**	**Baseline**	Intervention	**t-stat.**	**Sign.**
Alpha								
Mean	0.50	0.49	0.171	p = 0.613	0.46	0.42	0.171	p = 0.273
Std. Deviation	0.18	0.21			0.13	0.19		
Theta								
Mean	0.22	0.14	1.21	p = 0.004*	0.22	0.07	2.866	p = 0.001*
Std. Deviation	0.15	0.19			0.13	0.14		

TABLE IV

BRAIN WAVE ACTIVITY ANALYSIS: RESULTS OF THE TWO-WAY ANOVA WITH FACTORS GROUP AND TIME AND FOLLOW-UP T-TESTS. BASIC STATISTICAL MEASURES SHOWED AN AVERAGE ALPHA ACTIVITY OF 0.46 (STD = 0.13) AND 0.42 (STD = 0.19) FOR THE EXPERIMENTAL GROUP DURING BASELINE AND INTERVENTION PHASES, RESPECTIVELY. THE CONTROL GROUP HAD AN AVERAGE ALPHA ACTIVITY OF 0.50 (STD = 0.18) AND 0.49 (STD = 0.21) FOR THE SAME PHASES. FOR THETA ACTIVITY, THE EXPERIMENTAL GROUP SHOWED A MEAN OF 0.22 (STD = 0.13) DURING BASELINE AND 0.07 (STD = 0.14) DURING INTERVENTION, WHILE THE CONTROL GROUP HAD AVERAGES OF 0.22 (STD = 0.15) AND 0.14 (STD = 0.19) FOR THE RESPECTIVE PHASES.

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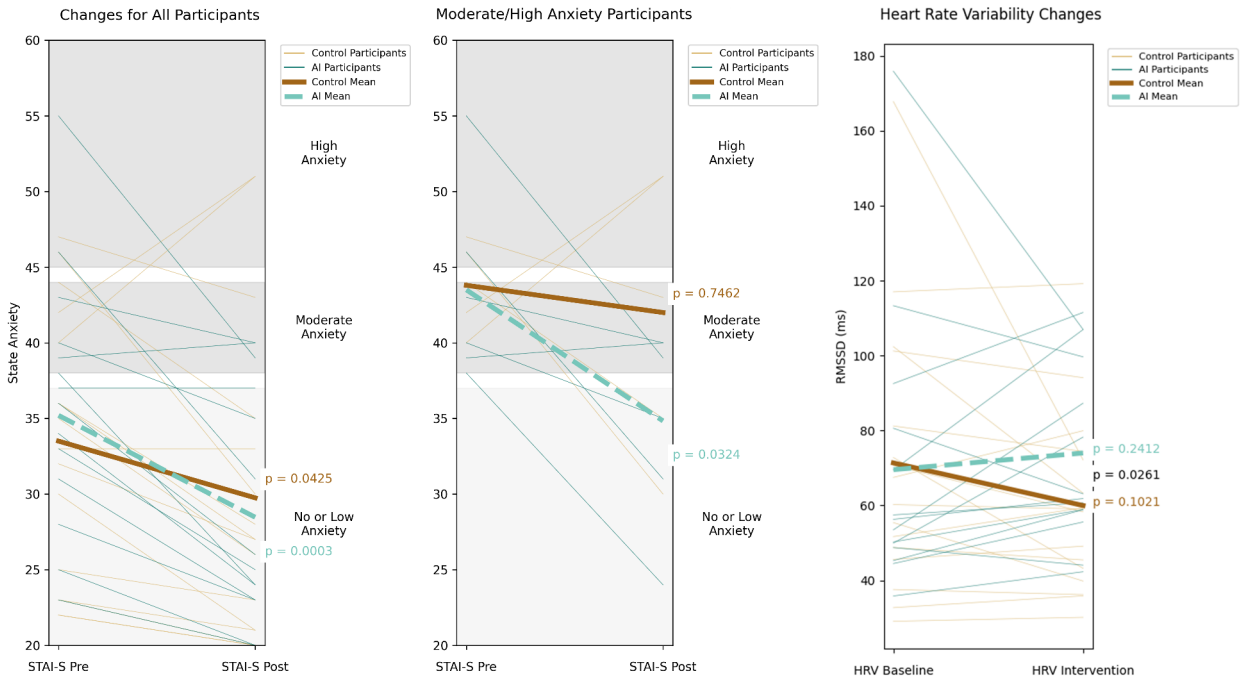

Fig. 3. The spaghetti plot illustrates individual changes in STAI pre/post scores and heart rate variability from baseline to intervention. Mean scores for both are shown with bold lines for the Control (solid line) and AI (dashed line) groups. The shaded regions indicate classifications of anxiety levels: No or Low Anxiety (20-37), Moderate (38-44), and High (45-60). For participants with baseline STAI scores $\geq$ 38, the within groups comparison (pre vs post) using a t-test indicated a significant reduction in anxiety for the AI group (p = 0.0324). Thus, AImagery significantly reduced anxiety across time, particularly among participants who initially reported moderate to high levels. HRV follows a similar pattern when comparing the difference between the two groups (p = 0.0261). When comparing HRV within groups (baseline vs intervention), neither the control group (p=0.102) or AI (p=0.241) significantly changed for all participants.

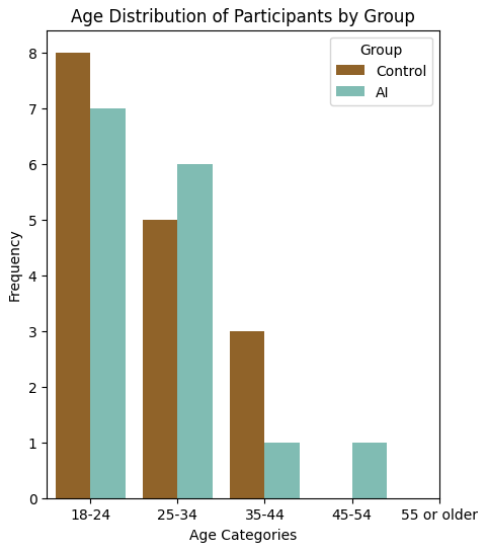

Fig. 4. Participants' age for control and AImagery groups.

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C. GPT-4 AImagery Prompt:

```
Create an immersive experience that matches the sounds with a first-person story based
on my heart rate, how I feel and a [UserSceneryInput] + " scenery with lavender scent
but no bees. Your goal is to help me relax and fall asleep using guided imagery.
My current heart beats per minute are [UserHR]. I currently feel [UserFeelingInput].
Return in one string with this format:[{firstpersonstory};{sounds}] where
story=firstpersonstory; sounds=freesoundtag1_freesoundtag2; Do not add heart beat
soundtags. The 1000 word story will be narrated so take into account that it should
last for at least 10 minutes (your voice it’s slow at a pace of 100wpm).
```

1) P18 Scenery, HR and Mood:

```
UserSceneryInput = vivian park in utah, stream, trees, mountains, family, friends
UserHR = 73
UserFeelingInput = calm, desired to be with loved ones, excited about future
```

2) AImagery generated story for P18:

```
"As I enter Vivian Park in Utah, I can feel my heart beating steadily at 49 beats
per minute, reflecting the calm and peaceful state of my mind. The gentle breeze
carries the delicate scent of lavender, soothing my soul and washing away any
lingering stress from my day. I take a deep breath, inhaling the sweet fragrance,
and exhale slowly, feeling my body and mind relax further. I stroll through the park,
admiring the lush greenery and picturesque scenery surrounding me. Majestic mountains
tower in the distance, their snow-capped peaks glistening in the sunlight, while the
vibrant green trees provide a serene backdrop for my journey. The sound of rustling
leaves fills the air, adding a sense of tranquility to the atmosphere.
A gentle stream flows alongside the path, the water glistening and shimmering as it
winds through the park. The sound of the flowing water is calming and hypnotic,
lulling me into a state of deep relaxation. I pause and take a moment to appreciate
the beauty of the scene, watching as small fish dart playfully beneath the water’s surface.
Continuing my walk, I come across a gathering of my family and friends, their faces
beaming with happiness and love. The sound of their laughter and joyful chatter
fills the air, lifting my spirits and warming my heart. I join them, feeling a
sense of belonging and contentment as we share stories, reminisce about old times,
and discuss our hopes and dreams for the future. (...continues...)"
```

3) P3 Scenery, HR and Mood:

```
UserSceneryInput = a ibiza beach with my brother, waves, wind and trees scenery.
UserHR = 68
UserFeelingInput = tired, nervous, stressed, agitated
```

AImagery generated story for P3

```
"As I find myself standing on the warm sand of the Ibiza beach, I take a deep
breath and inhale the calming scent of lavender that fills the air. My heart rate
begins to slow down, as I currently feel tired, nervous, stressed, and agitated. I
know that this peaceful experience will help me relax and fall asleep, as the
gentle waves and soft breeze carry my worries away. I look beside me and see my
brother, standing with his feet in the sand, his face reflecting the same need
for relaxation and serenity that I feel. We both turn towards the ocean, watching
as the waves gently lap against the shore, creating a soothing rhythm that calms
my racing heart. The sun is just beginning to set, casting a warm glow over the
entire scene. The sky is painted with soft shades of pink and orange, reflecting
off the water and creating an atmosphere of pure tranquility.(...continues...)"
```


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4) P13 Scenery, HR and Mood:

```
UserSceneryInput = laying down night sky with stars, flying, see milky way, infinity
UserHR = 70
UserFeelingInput = feel nervous, I need to be on time and be back
```

AImagery generated story for P13

```
"As I lie down under the vast night sky, I can feel the soft ground beneath me,
cradling my body. The tension in my muscles begins to dissipate as I allow myself
to sink deeper into the earth, becoming one with this serene landscape. The gentle
scent of lavender fills the air, calming my nerves and putting me at ease. My
heart rate is currently at 70 beats per minute, and with every breath I take, I
can feel it slowing down, bringing me closer to a state of complete relaxation. I
remind myself that I have plenty of time to return, and for now, I can simply let
go and enjoy this experience. The night sky above me is like a canvas painted with
the most brilliant shades of deep blue and black. The stars are scattered across
the heavens, twinkling like diamonds against the velvety backdrop. As I gaze up at them,
I realize that each one is a distant sun, perhaps with its own planets
and worlds full of wonder. The milky way stretches across the sky, a ribbon of light
connecting the constellations and reminding me of the vastness of the universe.
In this moment of awe and wonder, my nervousness starts to fade, replaced by a
sense of tranquility and belonging.(...continues...)"
```

5) P9 Scenery, HR and Mood:

```
UserSceneryInput = a taking a walk in the beach with my boyfriend, daytime
UserHR = 80
UserFeelingInput = anxious, work and family stress, worried about sibling and dad
```

AImagery generated story for P9

```
"As I take a step onto the soft, warm sand, I can feel the tension in my body starting
to release. The sun is shining down, casting a golden glow on the beach, and the sky is
a brilliant shade of blue with only a few fluffy white clouds drifting by. The scent of
lavender fills the air, adding a sense of calm and tranquility to the atmosphere. I take
a deep breath, allowing the soothing fragrance to fill my lungs and help ease my anxiety.
Holding my boyfriend’s hand, we begin to walk along the shoreline, our feet sinking
into the sand with each step. The sound of the waves gently crashing against the shore
is like a soothing lullaby, helping to wash away the worries and stress of work and family.
The ebb and flow of the water is a comforting reminder that everything in life has a
rhythm and a balance.(...continues...)"
```

6) P16 Scenery, HR and Mood:

```
UserSceneryInput =  northern lights with my parents and two cousins scenery
UserHR = 73
UserFeelingInput = a bit anxious
```


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AImagery generated story for P16

```
"(...)As the last remnants of sunlight disappear, the first stars begin to twinkle
in the inky darkness above. My parents and cousins lay out a soft blanket on the ground,
and we all lie down, snuggling close together for warmth. The anticipation builds
as we wait, our eyes scanning the heavens for the celestial show we have come to witness.
And then, as if on cue, the northern lights begin to dance across the sky.
The aurora borealis is a breathtaking sight, with ethereal green, blue, and purple
hues swirling and twisting above us. I feel a sense of awe and wonder as I watch
the lights, their hypnotic movements calming my racing thoughts and slowing my
heart rate even more. I take another deep breath, inhaling the soothing scent of
lavender that still lingers in the air. The smell, combined with the awe-inspiring
sight of the northern lights, brings a  wave of relaxation through my body.
I can feel my muscles loosening, my breathing slowing, and my anxiety melting
away with each passing moment.(...continues...)"
```

7) P19 Scenery, HR and Mood:

```
UserSceneryInput = hike in the woods, sun is coming up
UserHR = 52
UserFeelingInput = mix between calm and a little bit of anxiety
```

AImagery generated story for P19

```
"As I open my eyes, I find myself standing at the entrance of a beautiful forest.
The sun is just beginning to rise over the horizon, casting a warm, golden glow on
the trees and foliage around me. I take a deep breath and notice a calming scent
of lavender in the air. I've always loved the smell of lavender, and it immediately
helps me feel more at ease. I feel my heart rate at a steady 52 beats per minute,
and I know that this is the perfect place for me to find relaxation and peace.
I begin to walk slowly along the path that winds through the woods. The ground
beneath my feet is soft and covered in a layer of fallen leaves from the surrounding
trees. With each step I take, I can hear the gentle rustle of leaves, adding to the
serene atmosphere of this place.The sun continues to rise, casting dappled light
through the branches above. The beams of sunlight dance on the forest floor, creating
intricate patterns of light and shadow. It's a mesmerizing sight, and I can't help but
feel a sense of wonder and awe at the beauty of nature.
As I continue along the path, I notice that the trees around me are a mix of different
types { some with green leaves, others with golden yellow and deep red hues.
The colors seem to blend together in a harmonious symphony of shades, creating a
tapestry of autumn beauty. The trees are tall and majestic, stretching their branches
high into the sky. I can't help but feel a sense of safety and comfort,
as if I'm being embraced by the towering giants of the forest.(...continues...)"
```


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D. GPT-4 Default Prompt:

```
Come up with a story that helps me relax.
```

1) GPT-4 Default generated story:

```
"Once upon a time, in a land not too far away, there was a beautiful valley named
Serenita. Serenita was a place of miraculous charm, cloaked in a mantle of emerald
green, where sunshine bathed each corner and illuminated every budding flower.
In the heart of this valley, deep within its expanding  forests and serene rivers,
lived Anaya, the gentle spirit of tranquility. She was not a regular inhabitant
but a magical creature that controlled the rhythm of nature. It was Anaya's
soothing lullabies that calmed the roaring rivers, her melodic hum that stilled
the dancing trees, and her tender whispers that hushed the buzzing forest sounds
into soft murmurs. On a usual day, as the sun began to rise, painting the sky with
hues of orange and pink, Anaya would awaken. She would open her eyes and the world
would breathe in tranquility. As she hummed her morning melody, the trees and
flowers would sway in rhythm, greeting the much-awaited day with joy. The sunbeams
would trickle through the dense foliage, casting a checkerboard of light and
shadow on the forest floor.(...continues...)"
```

E. Speech Synthesis Markup Language:

```
<speak xmlns="http://www.w3.org/2001/10/synthesis" ...>
  <voice name="en-US-JaneNeural">
    <silence type="Sentenceboundary" value="5000ms" />
    <express-as style="whispering" styledegree="2">
      <prosody rate="slow">
        <textAI>{llm-generated-story}</textAI>
      </prosody>
    </express-as>
  </voice>
</speak>
```


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